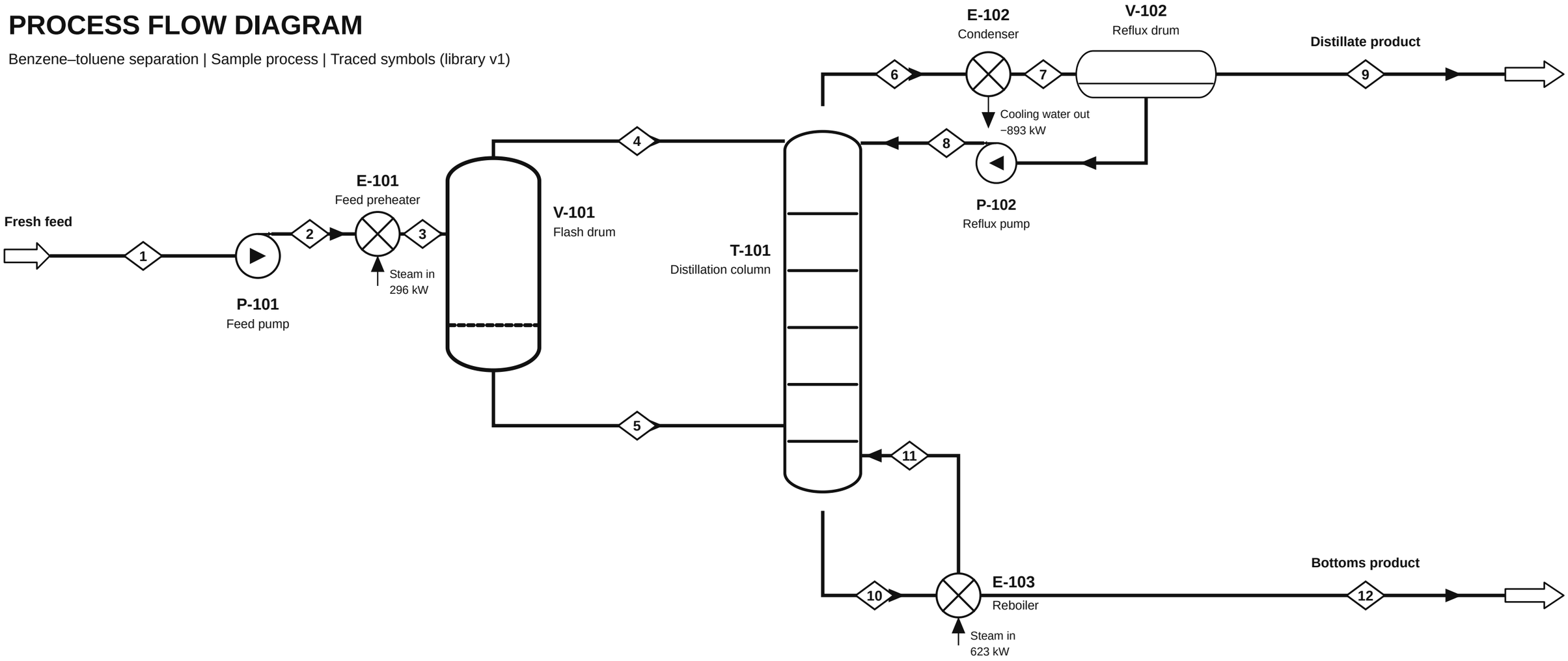
PROCESS FLOW DIAGRAM

Benzene–toluene separation | Sample process | Traced symbols (library v1)



Stream table — flow rates, temperatures, pressures and benzene contents are sample values

Stream	Flow rate (kmol/h)	Temperature (°C)	Pressure (kPa)	Benzene (mol %)	Phase
1	100.0	25	110	50.0	Liquid
2	100.0	26	150	50.0	Liquid
3	100.0	98	140	50.0	Two-phase
4	40.0	98	135	70.0	Vapour
5	60.0	98	135	36.7	Liquid
6	96.0	84	120	96.9	Vapour
7	96.0	82	115	96.9	Liquid
8	48.0	82	170	96.9	Liquid
9	48.0	82	115	96.9	Liquid
10	120.0	115	128	11.4	Liquid
11	68.0	115	125	15.0	Vapour
12	52.0	115	128	6.7	Liquid

Equipment summary

P-101	Feed pump	—
E-101	Feed preheater	296 kW
V-101	Flash drum	—
T-101	Distillation column	30 stages
E-102	Condenser (total)	−893 kW
V-102	Reflux drum	—
P-102	Reflux pump	Reflux ratio 1.0
E-103	Reboiler	623 kW

Benzene–toluene separation — PFD-101

Sample process flow diagram (layout demonstration)

Sample sheet | 2026-10-24 | Revision 1 | Sheet 1 of 1